

RV-Sparse: Coding Challenge

Challenge

Implement the `sparse_multiply` function that:

- Scans a row-major matrix A and identifies its non-zero elements.
- Extracts them into Compressed Sparse Row (CSR) format using caller-provided buffers.
- Computes the matrix-vector product $y = A * x$ using the extracted CSR data.
- Writes the result directly into a caller-provided output buffer.

Critical Constraint: Your function must perform zero dynamic memory allocation. All memory is pre-allocated by the caller.

Test Harness

The function prototype and test harness is given in the file `challenge.c` which you can find [here](#). The placeholder for the implementation of the function is also given there, marked by the **TODO** comment.

To test your implementation, execute the following.

```
>>> gcc -lm -o run challenge.c
>>> ./run
```

Submission

Host your solution in a public Git repository with brief documentation. Submit a PDF with the link to your Git repository.